

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION**Annual Examination 2026****CHEMISTRY – SSC-I (9th Grade)****Total Marks: 65****Time Allowed: 2 Hours 30 Minutes****Version: 1****General Instructions:**

1. Read all questions carefully before attempting.
2. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
3. **Section B** (Short Questions): Attempt **11 questions**, choosing either the main question **OR** its alternative. Each carries **3 marks**.
4. **Section C** (Long Questions): Attempt **4 questions**, choosing either the main question **OR** its alternative. Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.
6. No extra time will be given for Section A after 15 minutes.

SECTION A – Multiple Choice Questions (12 × 1 = 12 Marks)*Encircle the correct option. All questions are compulsory.*

#	Question	A	B	C	D	Answer
1	Chemistry deals with the composition, structure, properties, and changes of:	Energy	Space	Matter	Light	(A) (B) (C) (D)
2	The branch of chemistry that deals with carbon-containing substances (except carbonates, bicarbonates, oxides, carbides) is:	Inorganic	Physical	Organic	Analytical	(A) (B) (C) (D)
3	The branch that focuses on the study of polymers, their types, properties, and uses is:	Industrial	Polymer	Nuclear	Astro	(A) (B) (C) (D)
4	The integration of scientific knowledge for human needs is known as:	Science	Engineering	Technology	Research	(A) (B) (C) (D)
5	Forensic chemistry is the application of:	Organic	Analytical	Physical	Environmental	(A) (B) (C) (D)
6	Anything that has mass and occupies space is called:	Energy	Element	Compound	Matter	(A) (B) (C) (D)
7	A highly ionized gas where electrons are separated from atoms is called:	Solid	Liquid	BEC	Plasma	(A) (B) (C) (D)
8	Bose-Einstein Condensate (BEC) occurs at temperatures near:	100°C	0°C	-273.14°C	25°C	(A) (B) (C) (D)
9	Diamond, graphite, and buckyballs are allotropes of:	Silicon	Oxygen	Carbon	Nitrogen	(A) (B) (C) (D)
10	A solution containing maximum amount of solute at a given temperature is:	Unsaturated	Super-saturated	Dilute	Saturated	(A) (B) (C) (D)
11	The scattering of light by colloidal particles is called:	Reflection	Refraction	Tyndall effect	Dispersion	(A) (B) (C) (D)
12	Water is called a universal solvent because it:	Is colorless	Dissolves most compounds	Is everywhere	Has no taste	(A) (B) (C) (D)

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative from each pair.

Q.	Question	Marks		OR Question	Marks
2(i)	Define chemistry. List any four branches of chemistry with their definitions.	3	OR	What is green chemistry? Why is it important in reducing pollution?	3
2(ii)	Differentiate between organic chemistry and inorganic chemistry.	3	OR	Differentiate between geochemistry and astrochemistry.	3
2(iii)	Differentiate between science and technology with examples.	3	OR	Define engineering. How do chemical engineers contribute to industry?	3
2(iv)	Explain with an example how science, technology, and engineering work together to solve real-world problems.	3	OR	Describe two daily life applications of chemistry from different branches.	3
2(v)	Define matter. List the four commonly known states of matter with one example each.	3	OR	What is Bose-Einstein Condensate (BEC)? What are liquid crystals? Give one example of each.	3
2(vi)	Compare the macroscopic properties (density, compressibility, fluidity) of solids, liquids, and gases.	3	OR	Define plasma. Give three examples of plasma found in nature or daily life.	3
2(vii)	Define element, compound, and mixture with one example each.	3	OR	Differentiate between compound and mixture (any three differences).	3
2(viii)	What is allotropy? Name and briefly describe three allotropes of carbon.	3	OR	Describe the structure and properties of diamond. Why is it non-conductive?	3
2(ix)	Define solution, solute, and solvent. Give one example of each.	3	OR	What is an aqueous solution? Why is water called a universal solvent?	3
2(x)	Differentiate between saturated and unsaturated solutions.	3	OR	What is a supersaturated solution? How can it be prepared?	3
2(xi)	Differentiate between colloid and suspension (any three differences).	3	OR	Explain the effect of temperature on the solubility of solutes with examples.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt 4 questions. Choose either the main question OR its alternative from each pair.

Q.	Question	Marks		OR Question	Marks
3(i)	Describe any five branches of chemistry with their definitions and one daily life application of each.	5	OR	Explain the relationship between science, technology, and engineering with at least two detailed examples from the field of chemistry.	5
3(ii)	Describe the states of matter in detail. Include solid, liquid, gas, and plasma with their properties and examples.	5	OR	What are allotropes? Explain the three allotropes of carbon (diamond, graphite, and buckyballs) in detail including their structure and properties.	5
3(iii)	Differentiate between elements, compounds, and mixtures with examples. Explain the classification of matter with the help of a flowchart/diagram.	5	OR	What is a solution? Describe the types of solutions (gaseous, liquid, solid) with at least two examples of each type.	5
3(iv)	Explain saturated, unsaturated, and supersaturated solutions with examples. Discuss the effect of temperature on solubility with reference to the solubility table.	5	OR	Compare solutions, colloids, and suspensions in detail. Include particle size, visibility, filtration, Tyndall effect, and stability with examples.	5

Invigilator's Signature

Candidate's Signature

Chapters Covered: Ch 1 — Nature of Chemistry in Science | Ch 2 — Matter

— End of Question Paper —